Shahjalal University of Science and Technology, Sylhet.

Lab Report

Department of Electrical and Electronic Engineering

"Design and analysis of Rectangular Patch Antenna in CST Studio."

Course Code: **EEE 476**
Course Title: **RF Lab**
Experiment No.:**07**
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Submitted By

Name: Md. Takoat Hossain

Reg. No.: 2020338042

Group No:01

Submitted To

Md. Tasfiq Rahman,

Lecturer,

Department of Electrical and Electronic Engineering

Shahjalal University of Science and Technology, Sylhet.

Introduction

Microstrip patch antennas are widely utilized in modern wireless communication systems because of their low profile, ease of fabrication, and compatibility with microwave integrated circuits. Their lightweight, planar configuration makes them ideal for compact and efficient applications such as WLAN, GPS, and IoT systems. In this assignment, a rectangular microstrip patch antenna is designed to operate at a resonant frequency of 2.42 GHz, which lies within the S-band frequency range. The design, simulation, and optimization were carried out using CST Studio Suite. Various parameters including patch dimensions, substrate height, and dielectric constant were varied in parametric studies to evaluate their effects on antenna performance.

Objectives

- Design a rectangular microstrip patch antenna operating at 2.42 GHz using theoretical calculations.
- Perform parametric studies on substrate height and dielectric constant to observe their influence on resonance and bandwidth.
- Validate theoretical design equations with CST simulation outcomes.

Theory

The rectangular microstrip patch antenna primarily operates in the TM_{10} mode, where the electric field varies along the patch's length and remains nearly uniform across its width. The resonant frequency (f_r) for the dominant mode can be determined using:

$$f_r = c / (2 \times L_{eff} \times \sqrt{\varepsilon_{eff}})$$

Where:

- c = speed of light (3×10^8 m/s)
- L_{eff} = effective length of the patch
- ε_{eff} = effective dielectric constant

The patch width (W), effective dielectric constant ($\varepsilon_e$ff), and actual length (L) of the antenna can be determined using the following equations:

$$W = c / (2 \times f_r) \times \sqrt{(2 / (\varepsilon_r + 1))}$$

$$\varepsilon_{eff} = (\varepsilon_r + 1)/2 + (\varepsilon_r - 1)/2 \times [1 / \sqrt{(1 + 12h/W)}]$$

$$L = c / (2 \times f_r \times \sqrt{\varepsilon_{eff}}) - 0.824h \times ((\varepsilon_{eff} + 0.3) \times (W/h + 0.264)) / ((\varepsilon_{eff} - 0.258) \times (W/h + 0.8))$$

Calculation (for 2.42 GHz)

Given parameters:

- ε_r = 4.4
- h = 1.60 mm
- f_r = 2.42 GHz
- c = 3 × 10^8 m/s

Computed results:

- Patch Width (W) = 37.72 mm
- Effective Dielectric Constant ($\varepsilon_e$ff) = 4.084
- Length Extension (ΔL) = 1.477 mm
- Actual Patch Length (L) = 27.72 mm

- **Value of W and L from the website :**

Rectangular Microstrip Patch Antenna Calculator

Input

Resonant Frequency	f_r	2.42	GHz
Substrate Relative Permittivity	ϵ_r	4	
Substrate Height	h	1.5	millimeter

Output

Patch Physical Width	W	39.20179	millimeter
Patch Physical Length	L	30.63789	millimeter
Effective Length	L_{eff}	3.20433	centimeter
Input Impedance at Edge$(y=0)$	R_{in}	305.79094	ohm
50 ohm Feed Position	y_0	1.12592	centimeter
Single Slot Conductance	G_1	0.00104197	mho
Mutual Conductance	G_{12}	0.00059314	mho
Directivity	D	6.20054	dBi

Compute

Fig: Website Computed Parameters

- **Patch antenna Design:**

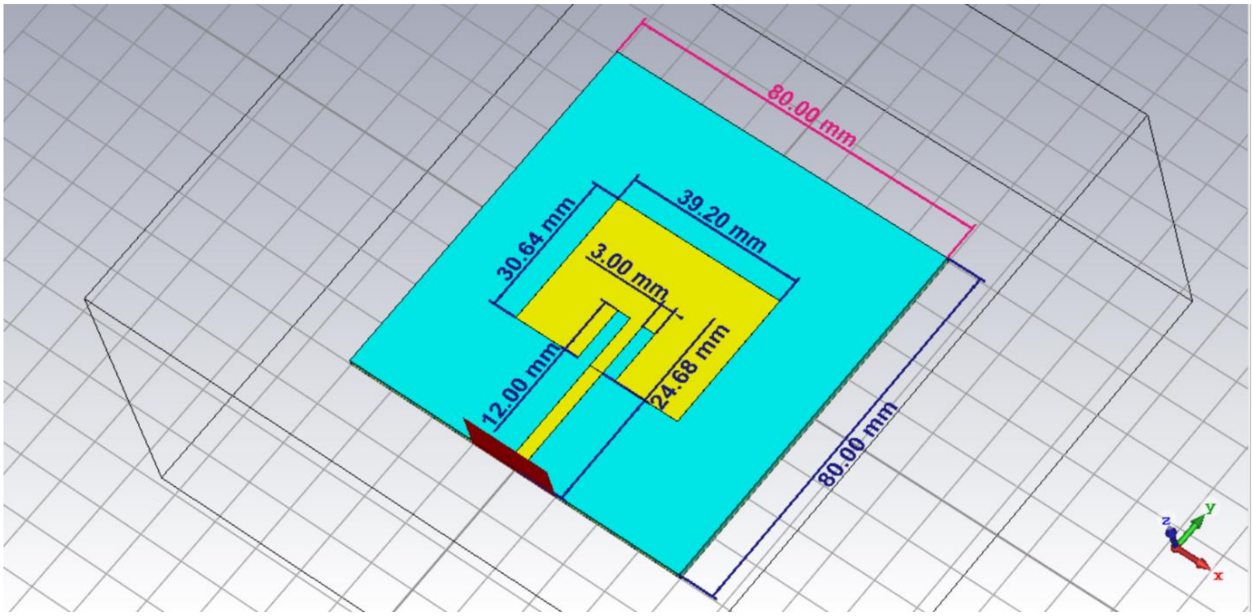


Fig: Rectangular Patch Antenna CST Design

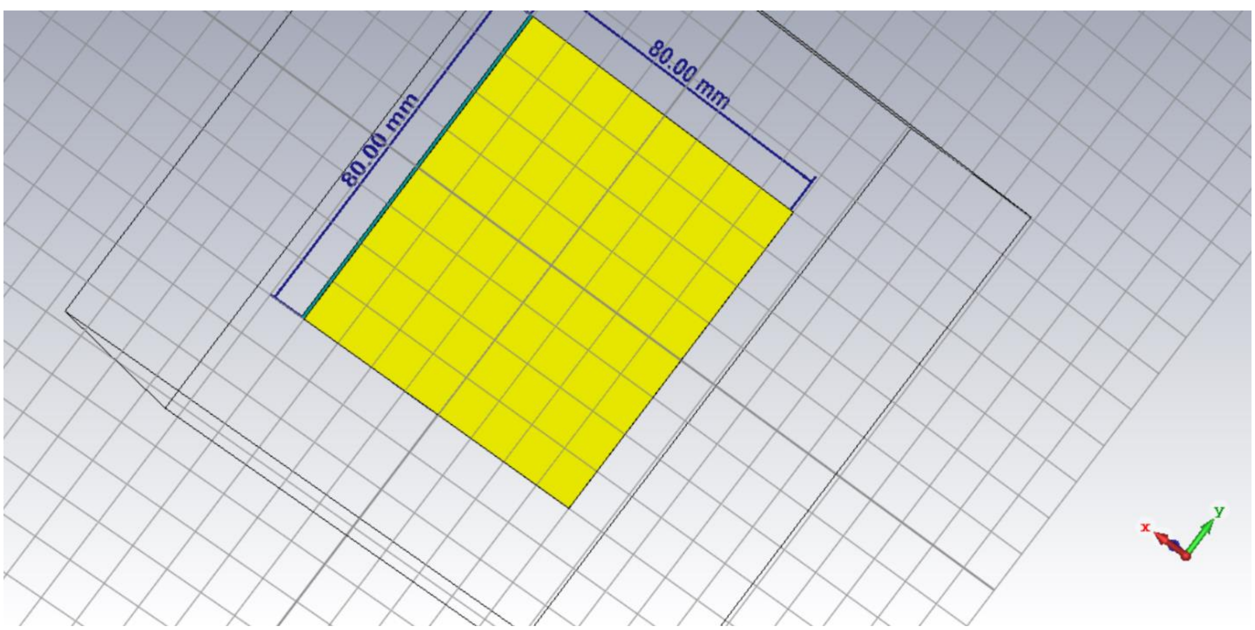


Fig: Antenna Back ground plane

- **Port Extension Co efficient calculate**

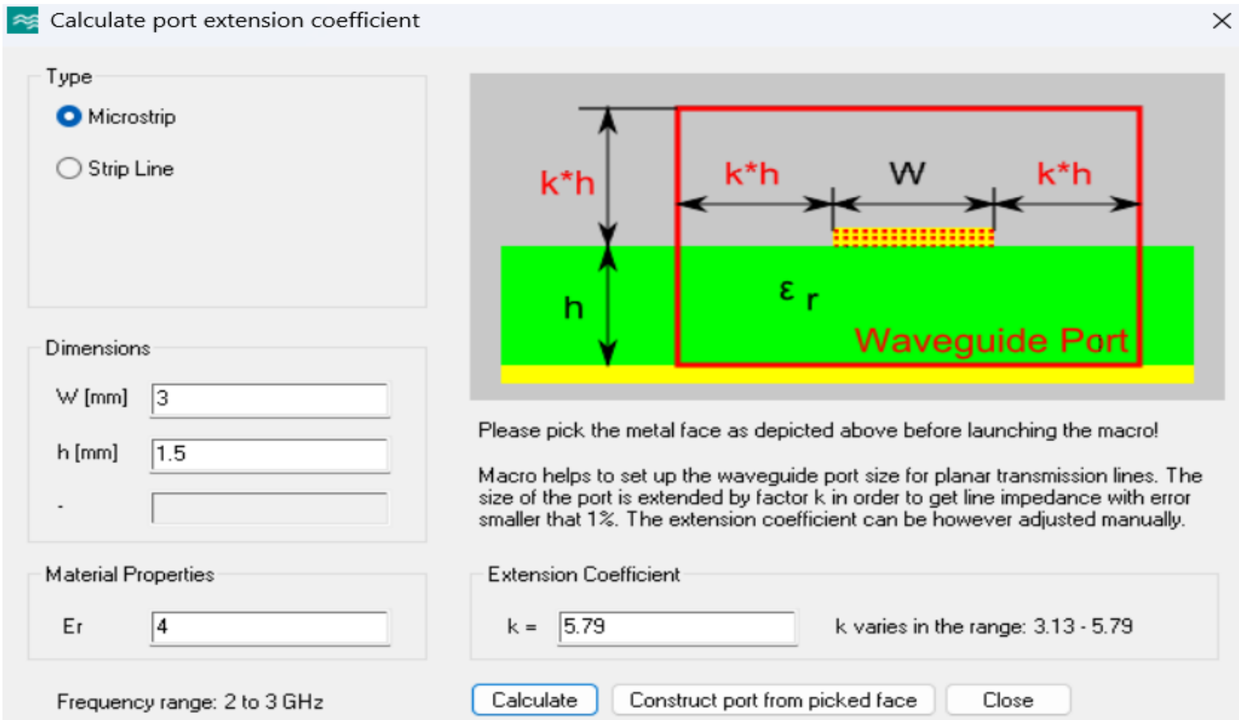


Fig: Port Extension Parameters

1. **Resonant frequency using the values obtained from the website and dielectric layer 1.5 and permittivity 4 at resonance frequency at 2.42 GHz :**

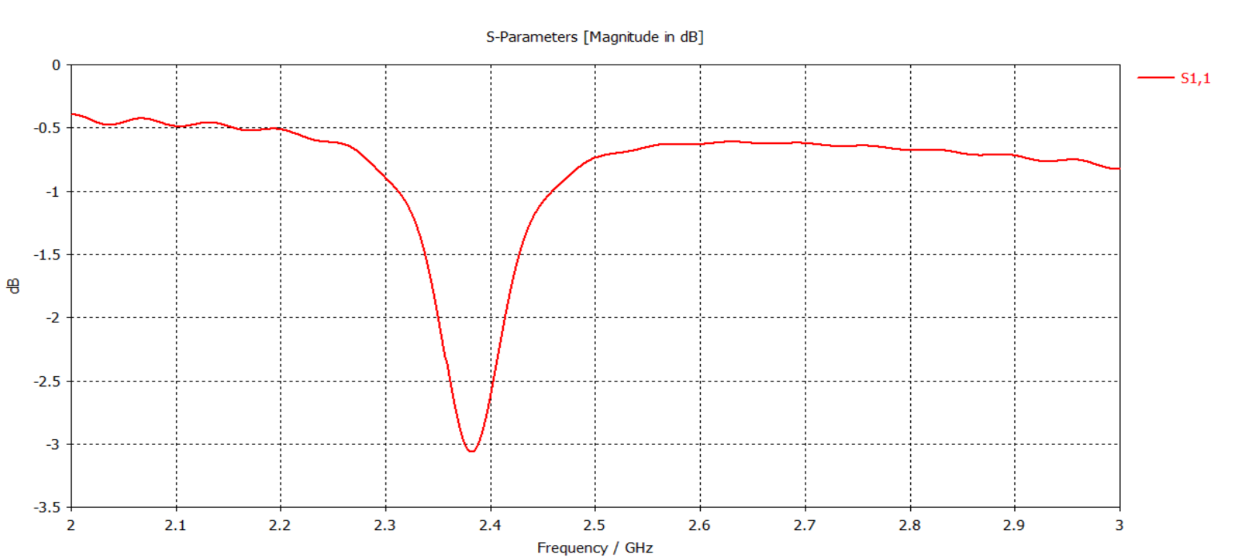


Fig: Resonant Frequency at 2.42 Ghz

- **Power :**

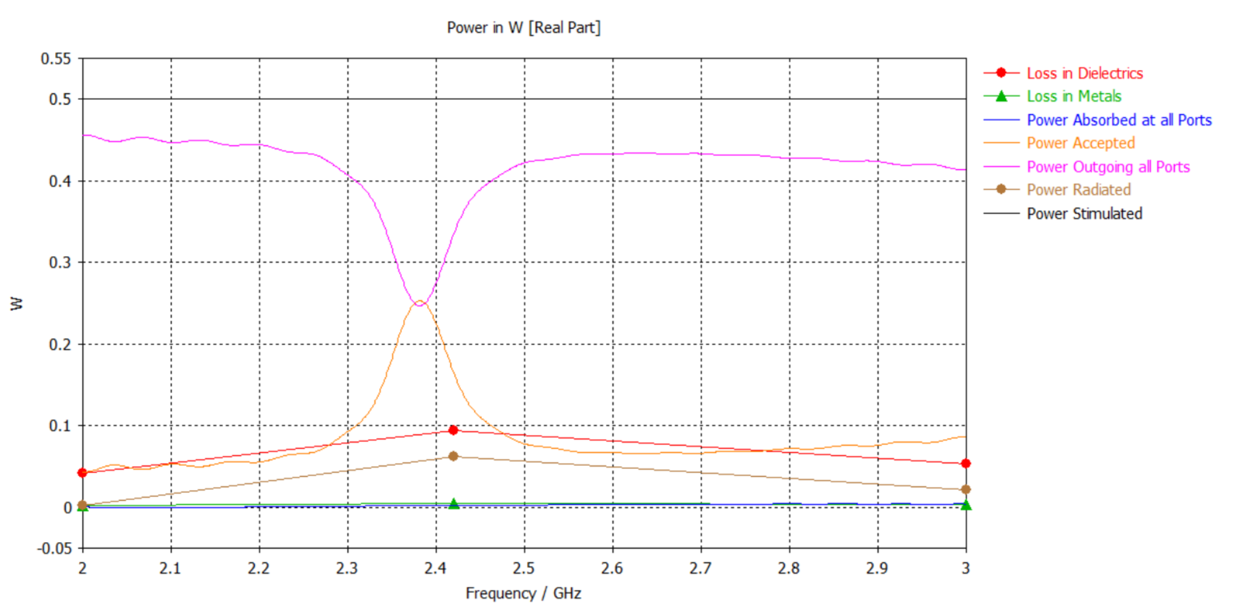


Fig: Power at 2.42 GHz Resonant Frequency

- **Farfield polar & 3D plot:**

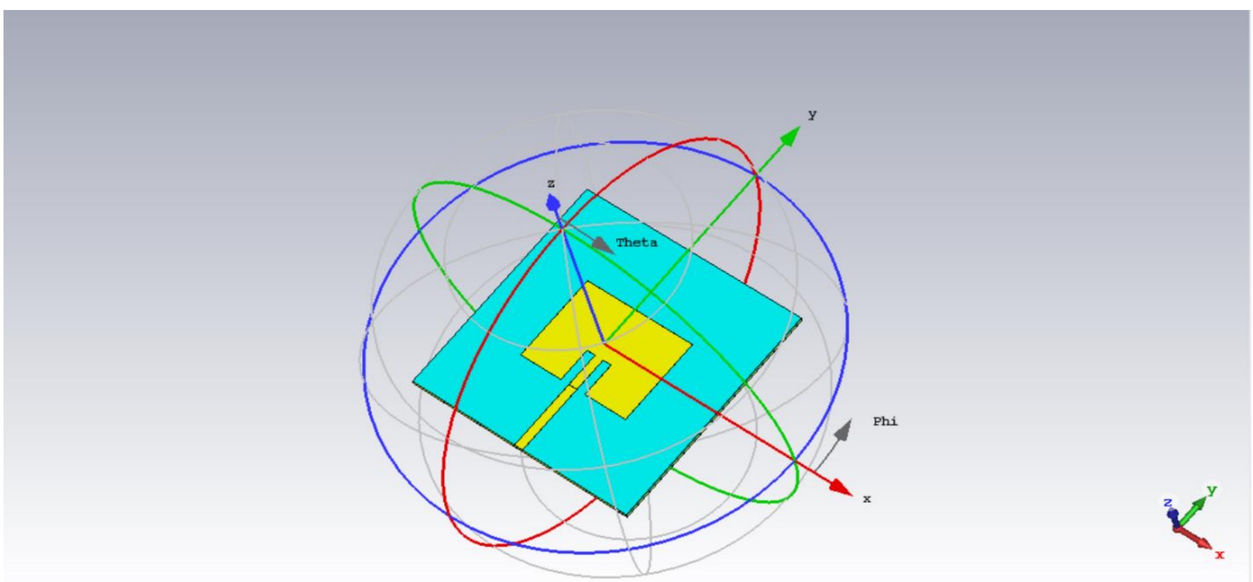


Fig: Farfield Orientation

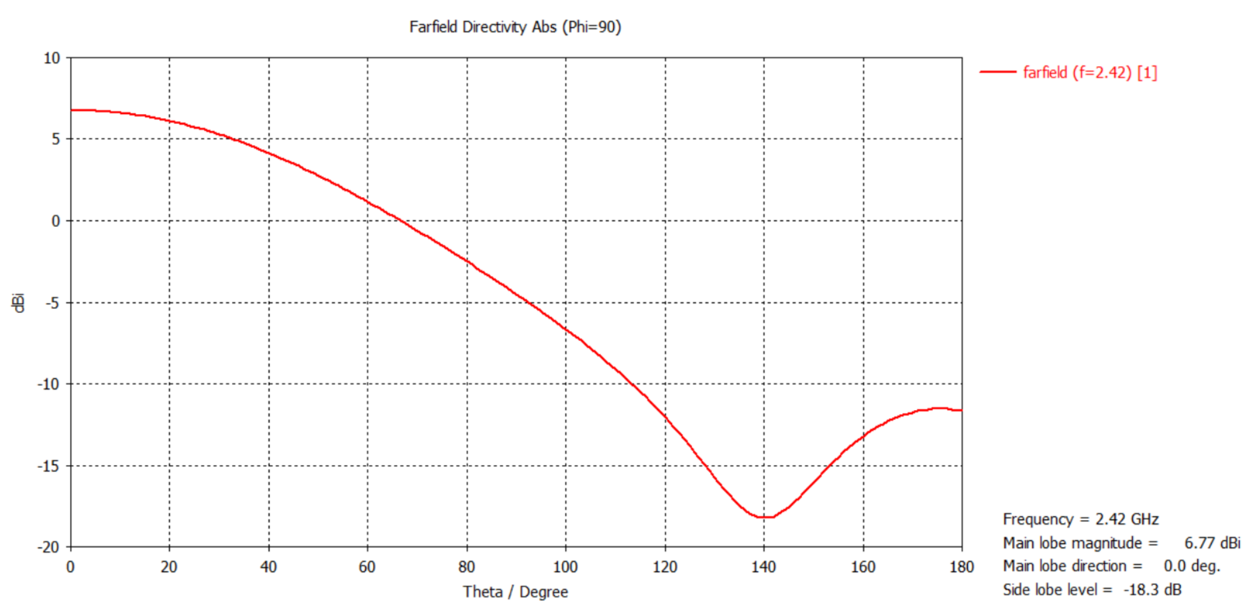


Fig : Farfield Directivity

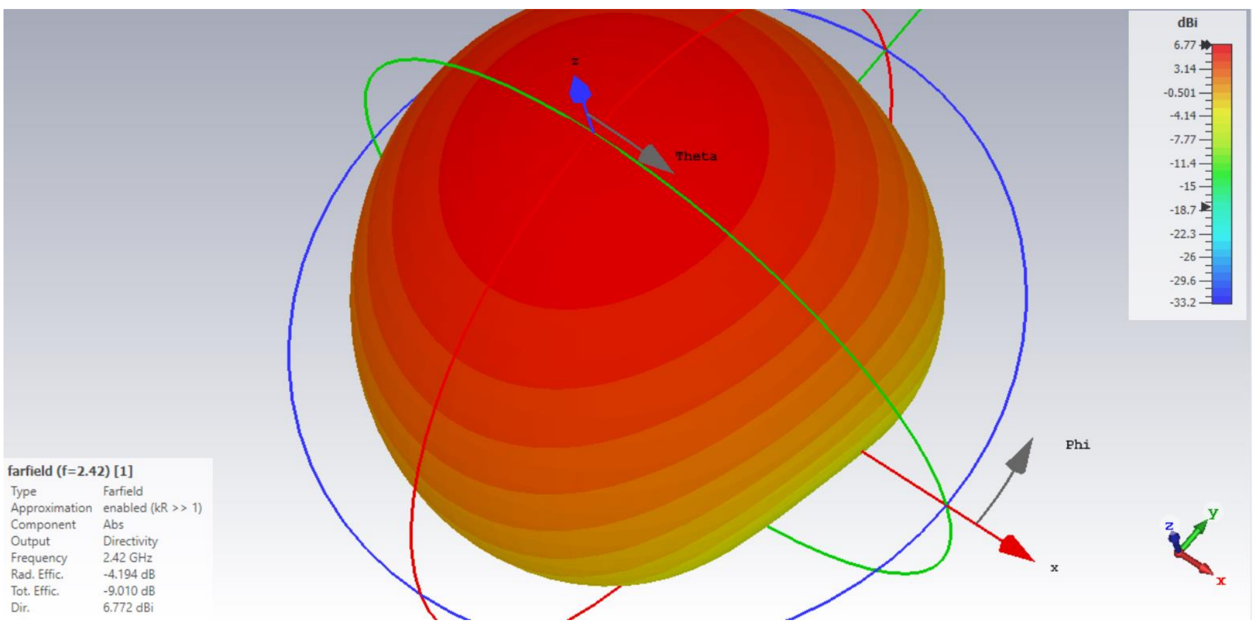


Fig: Farfield Radiation 3D Plot

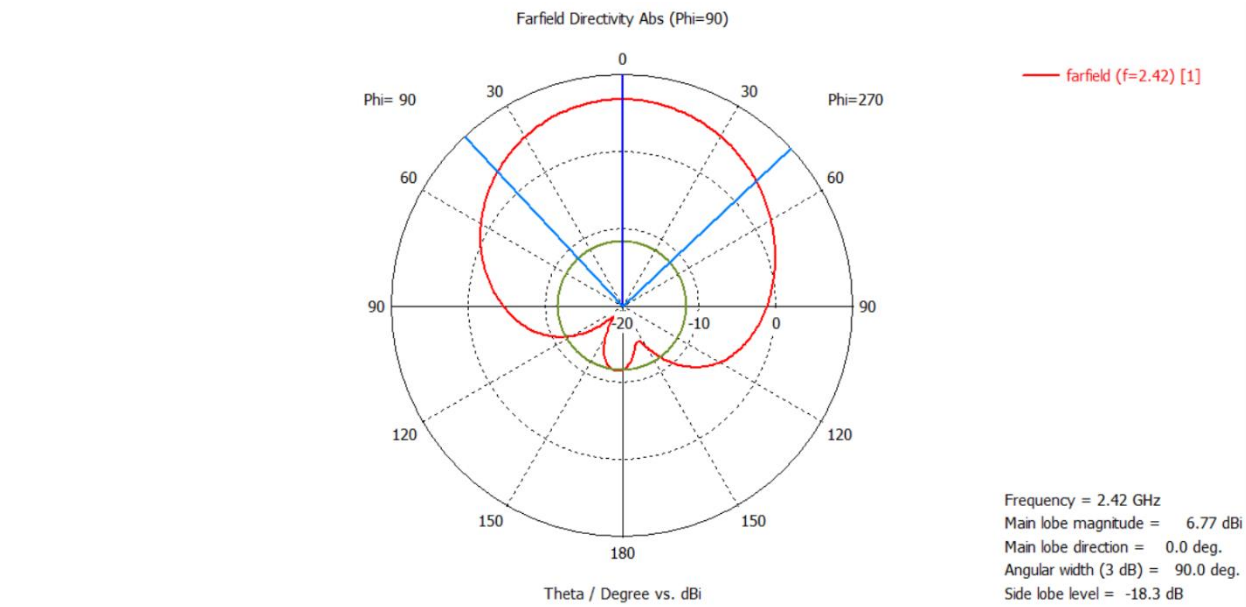


Fig: Farfield Directivity Polar Plot

2. Change in resonant frequency due to change in dielectric layer to 1mm and 2mm

- **1mm :**

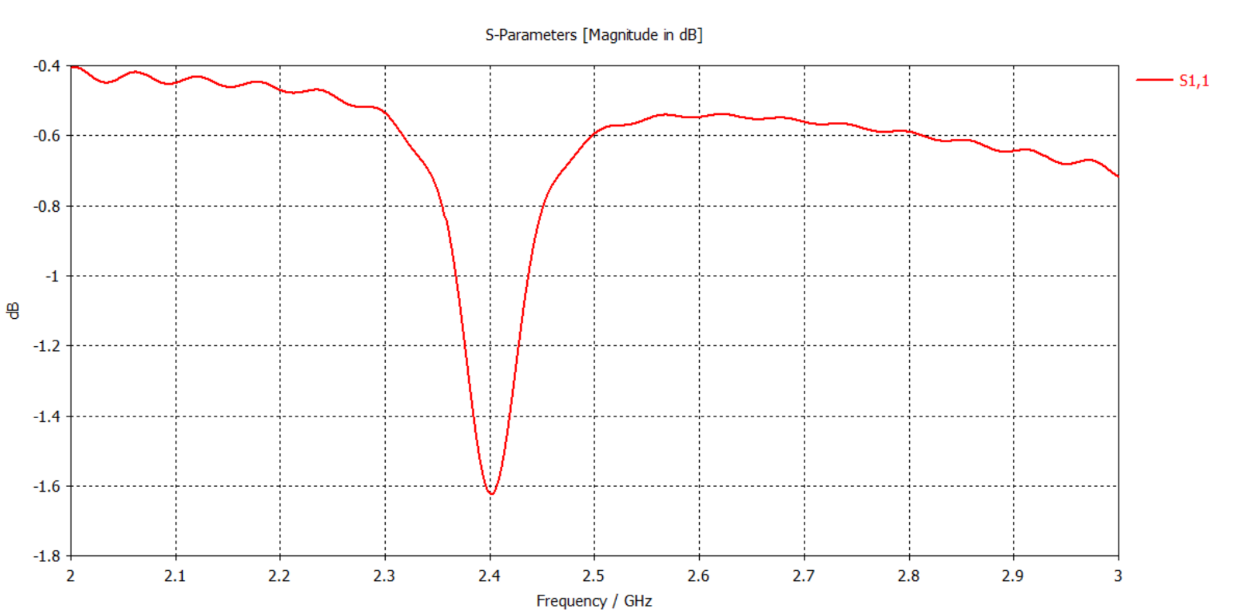


Fig: S-Parameter

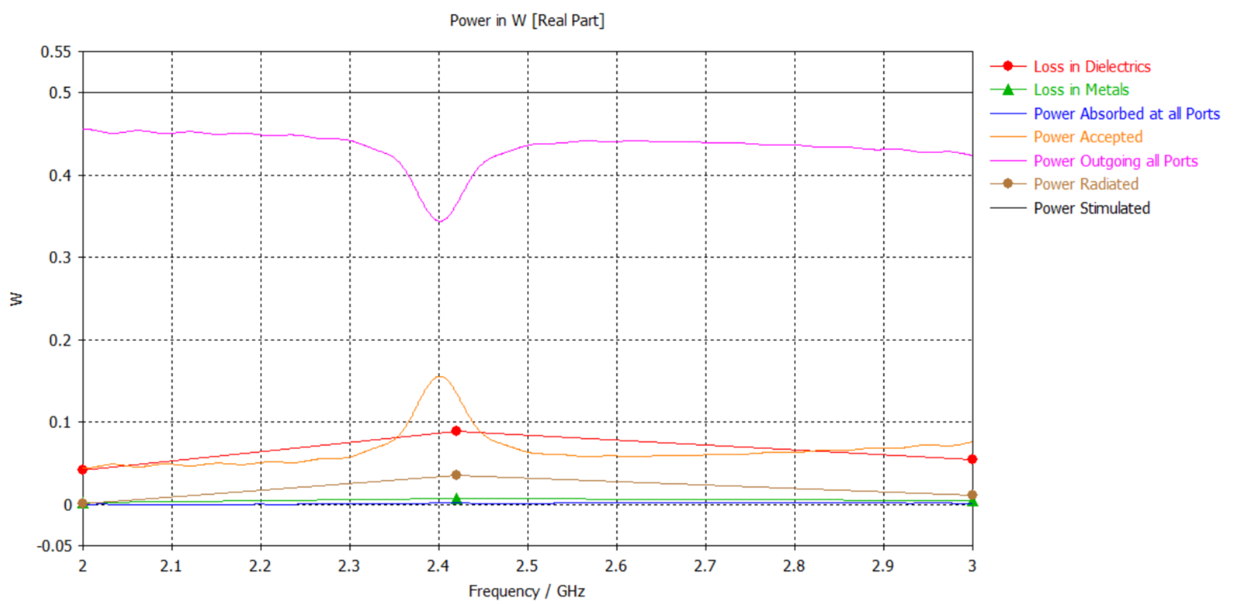


Fig: Power

- **2mm :**

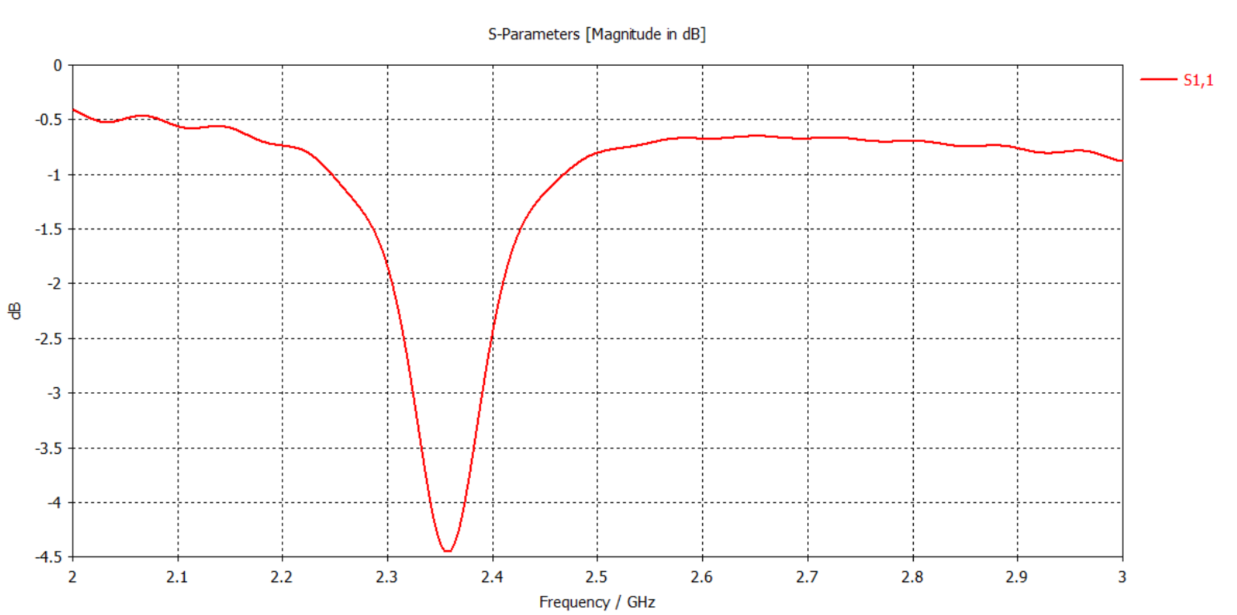


Fig: S-Parameter

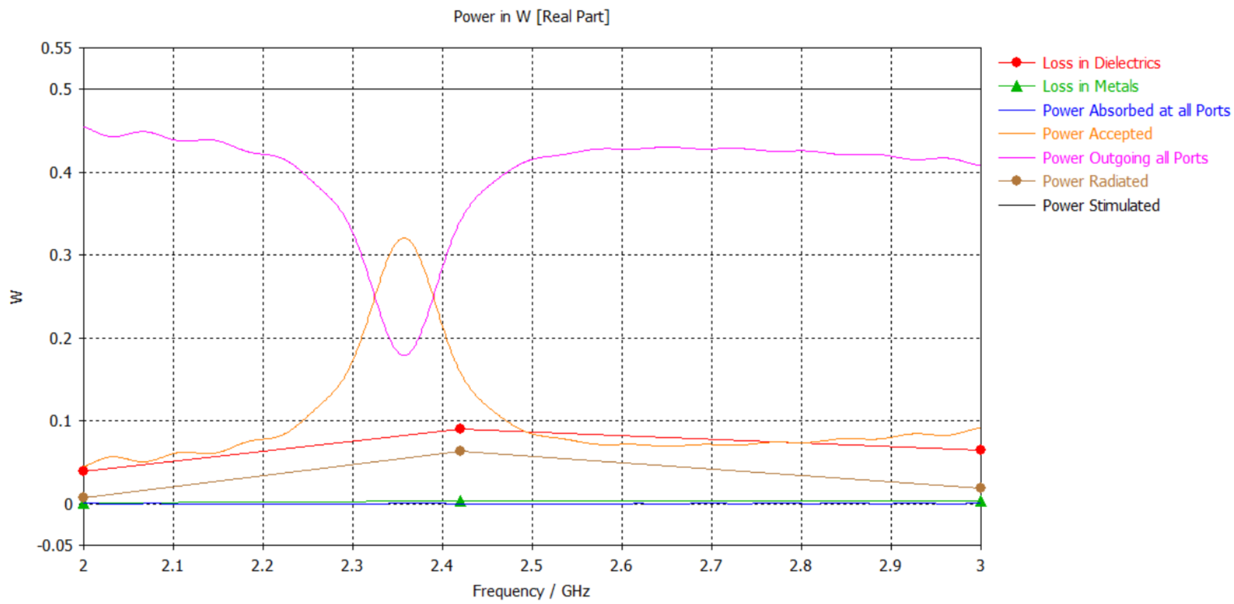


Fig: Power

3. Change in resonant frequency when di-electric constant is changed to 2 and 7:

- **For ε_r to 2**

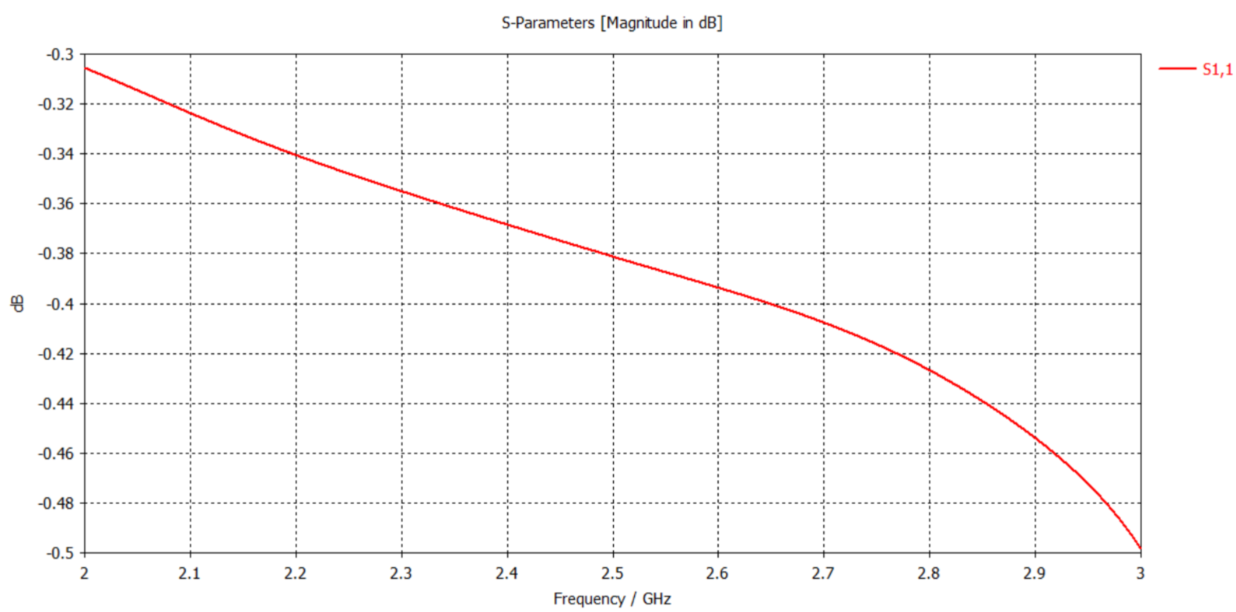


Fig: S-Parameter

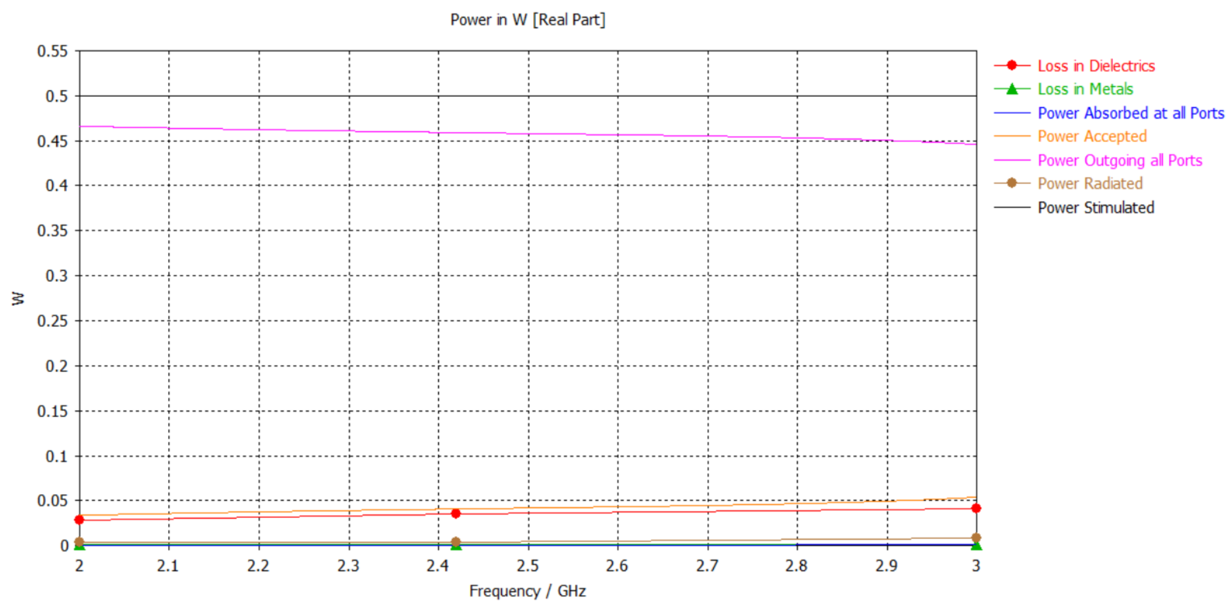
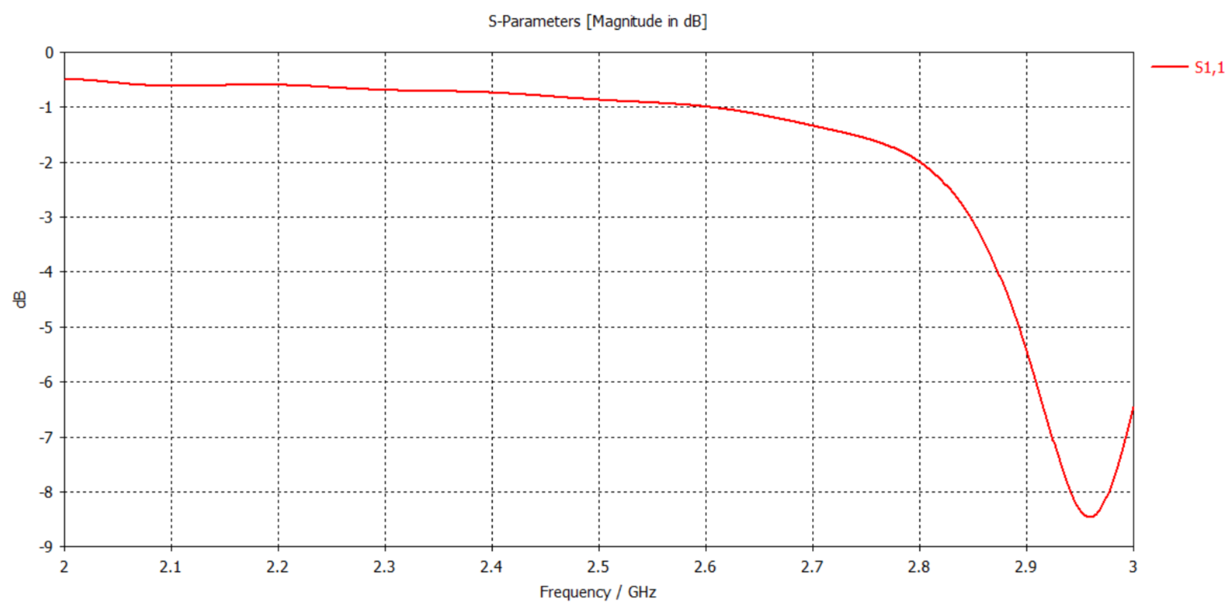


Fig: Power

- **For ε_r to 7**



Fig; S-Parameter

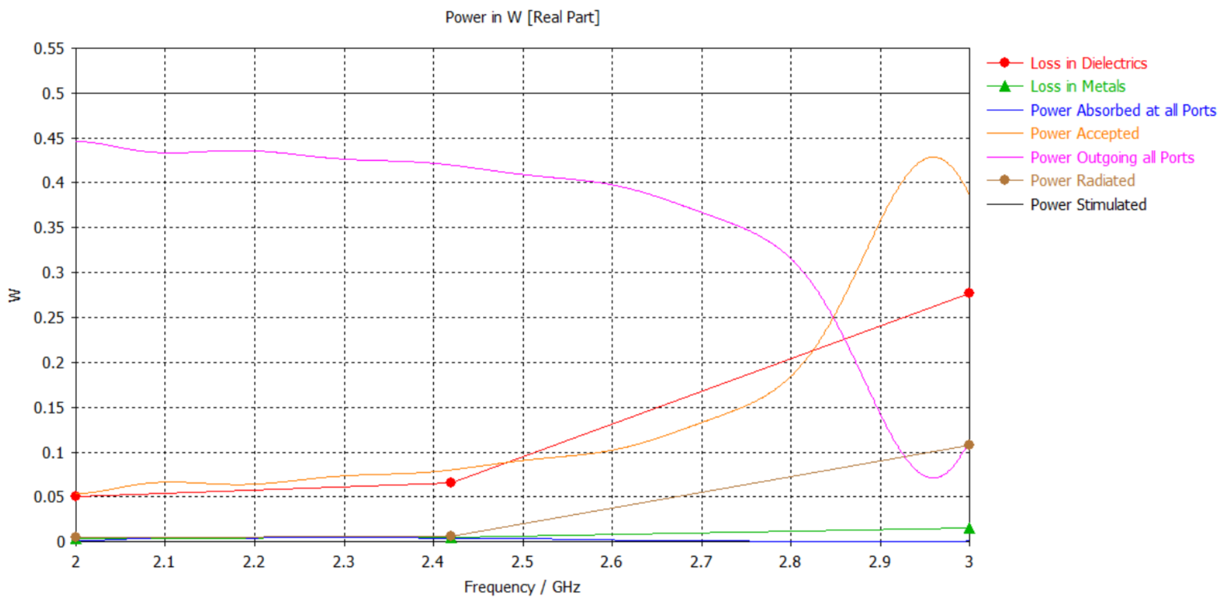


Fig: Power

4. **changing 0.035mm thickness to GND antenna tx line layer to 0.5mm and 0.001mm and resonant frequency change and polar and 3D far field power plot:**

- **For 0.5 mm:**

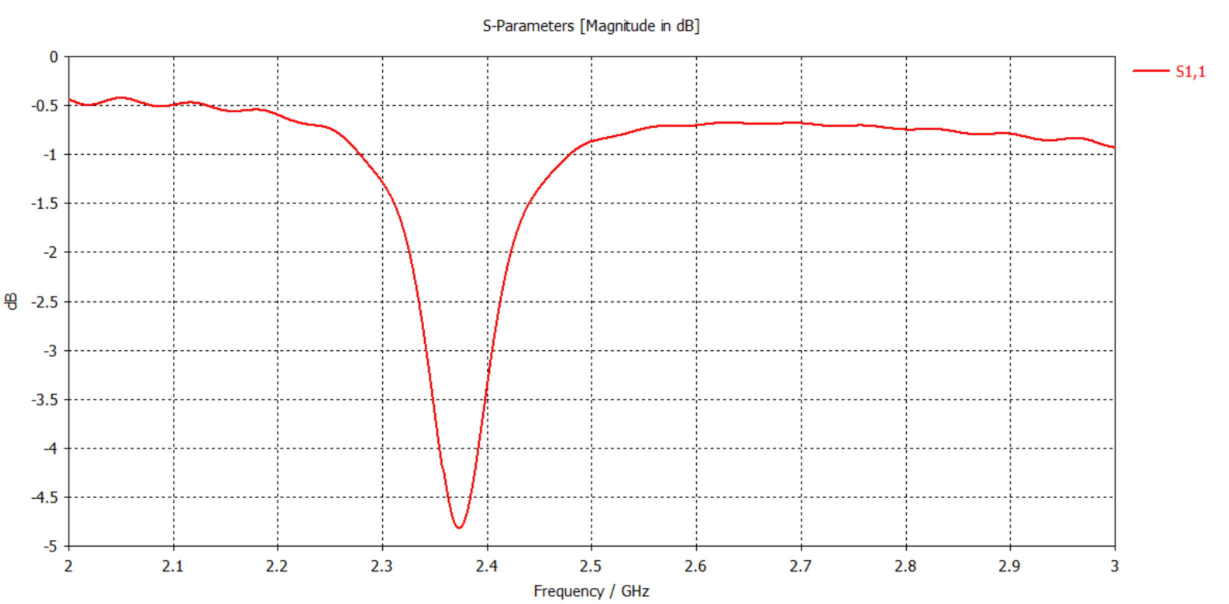


Fig: S-Parameter

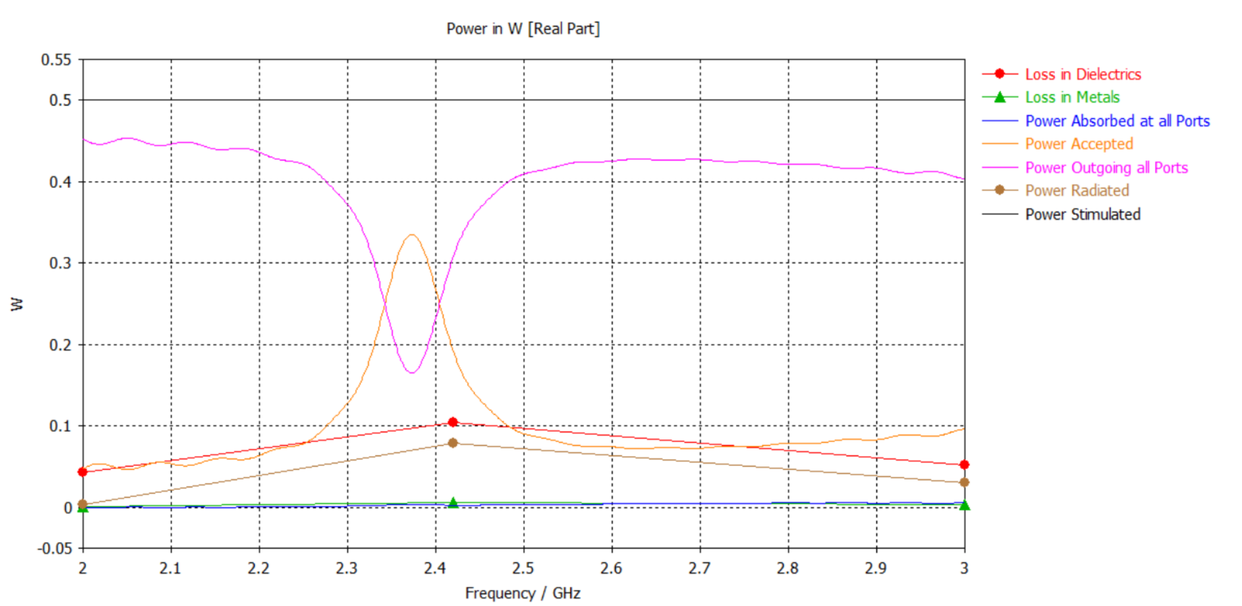


Fig: Power

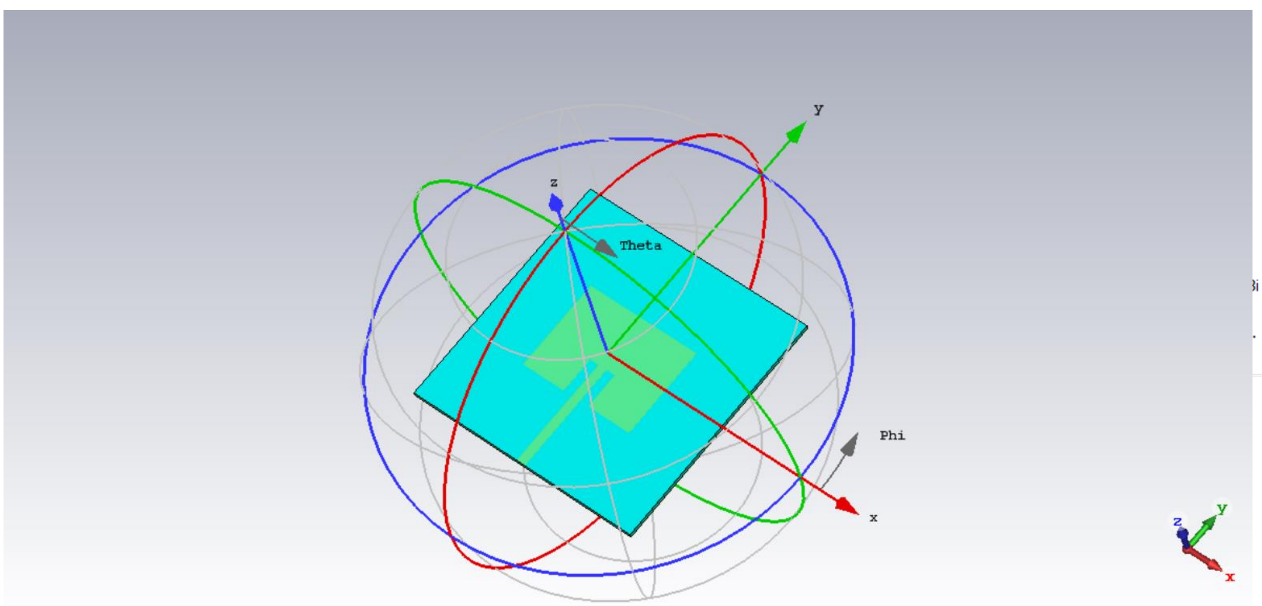


Fig: Farfield Orientation

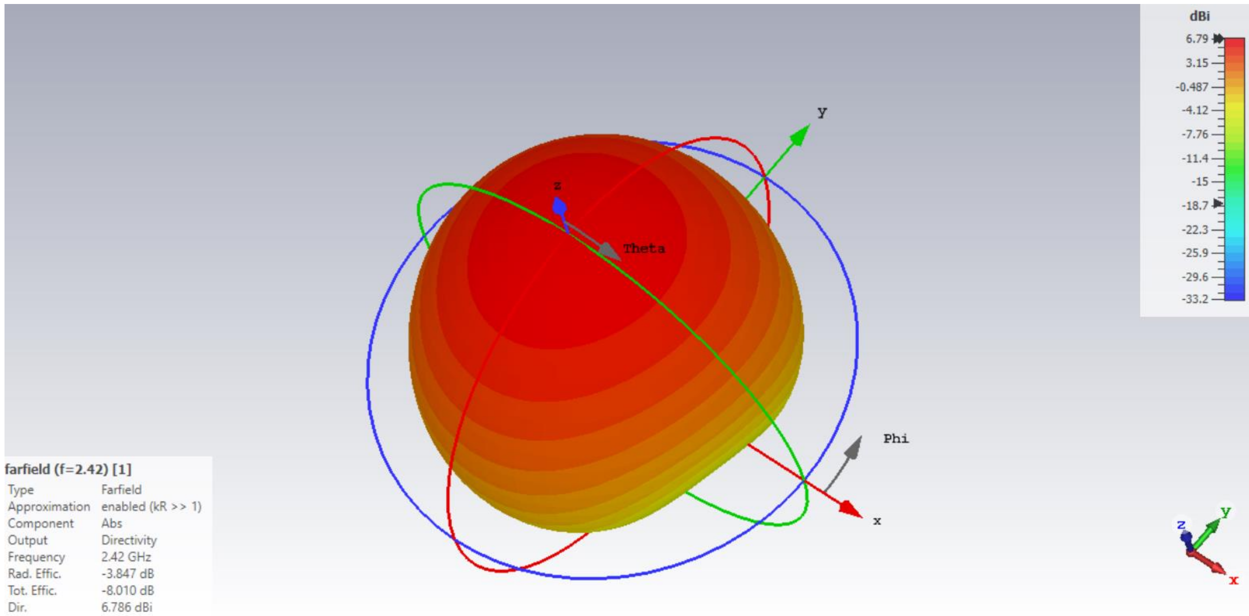


Fig: Farfield Directivity in 3D

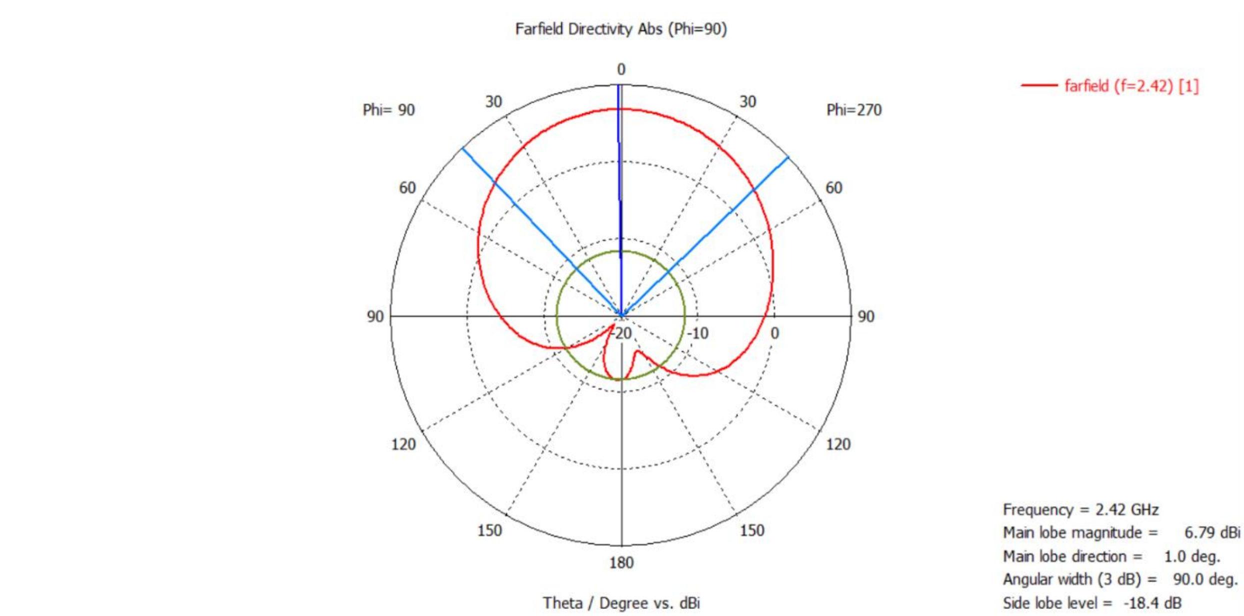
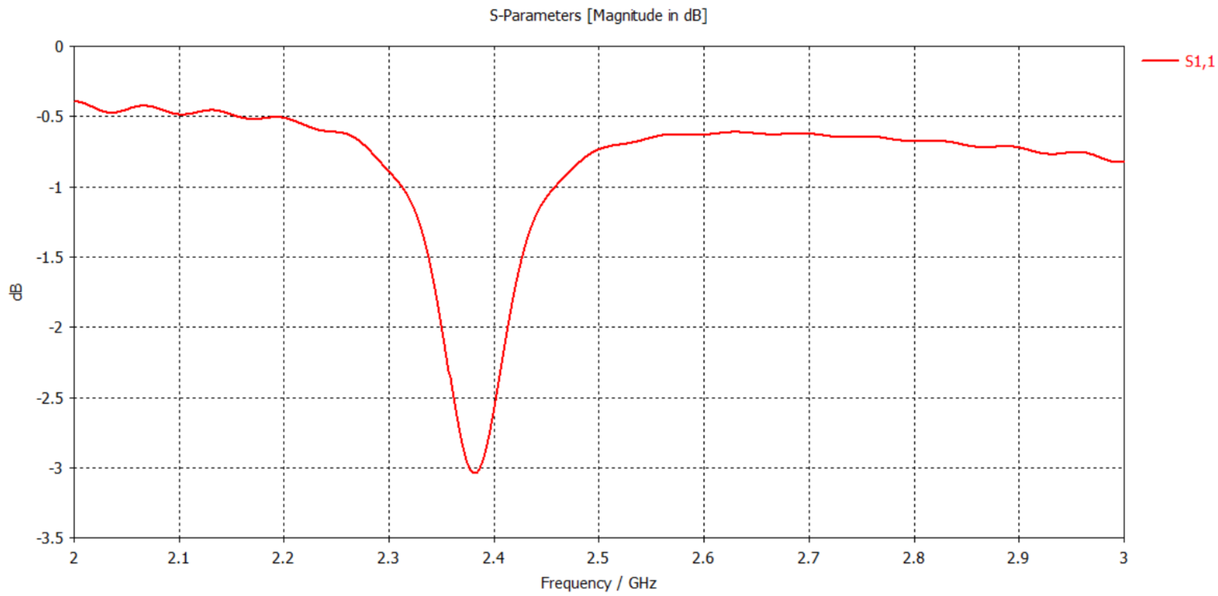


Fig: Directivity

- **For 0.001 mm:**



Fig; S-Parameter

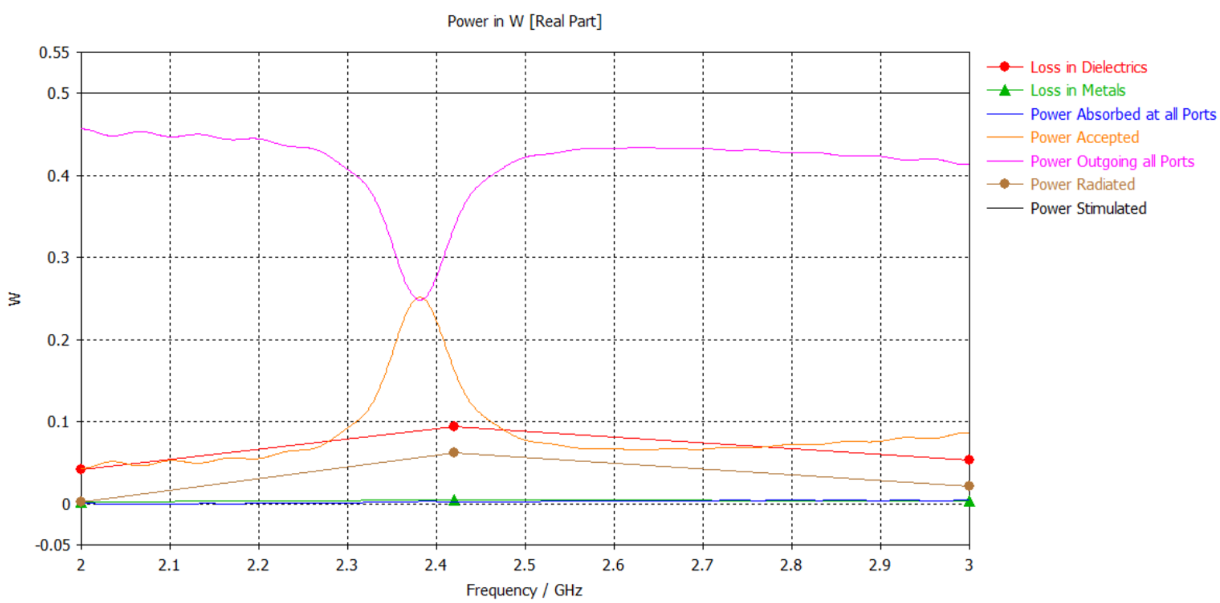


Fig: Power

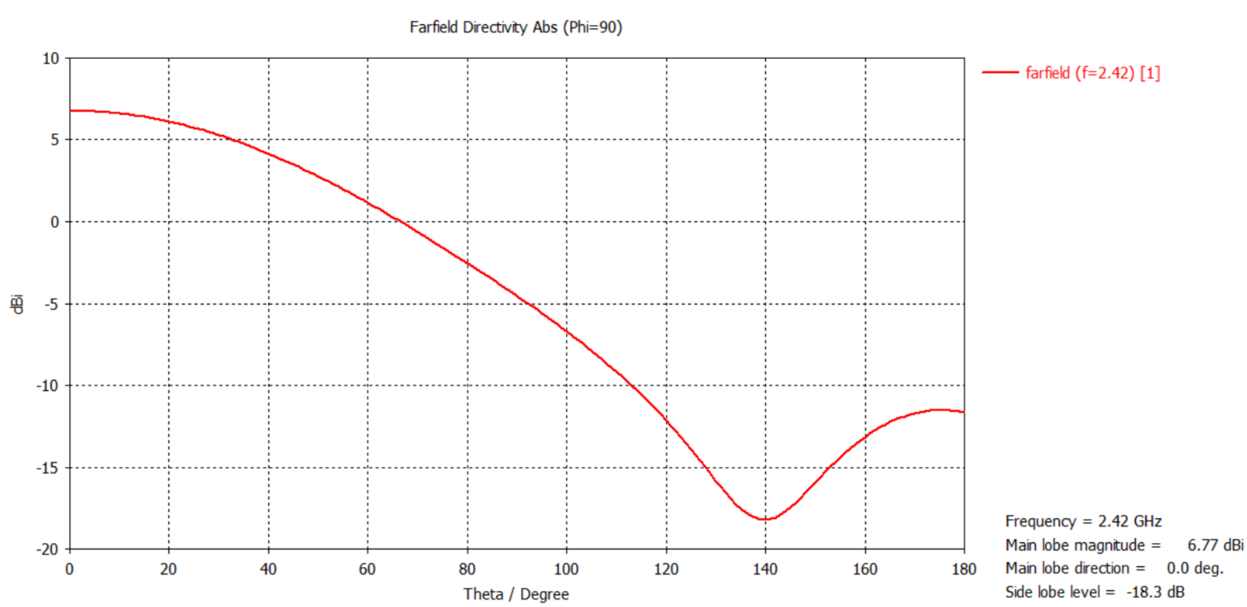


Fig: Field Directivity

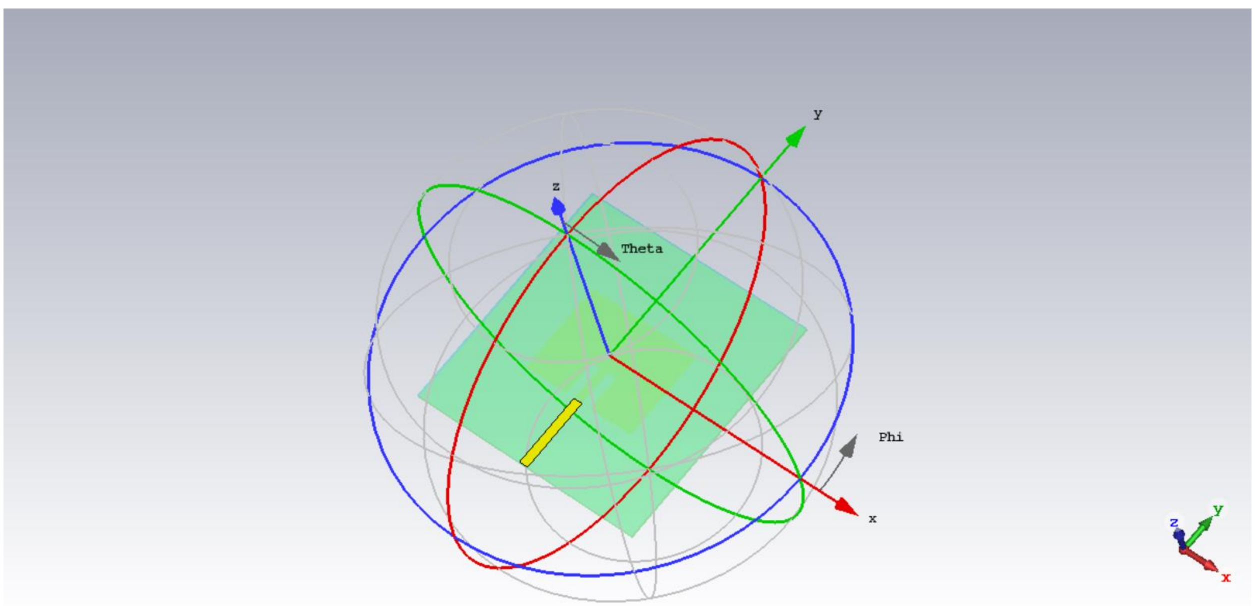


Fig: Farfield Orientation

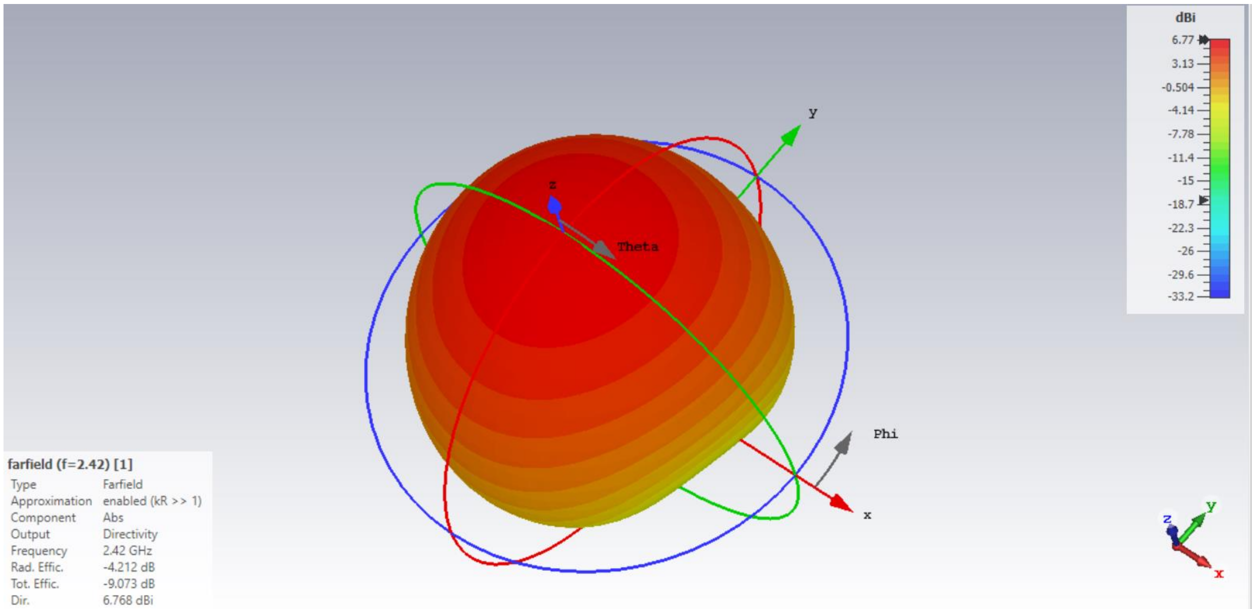


Fig: Farfield Directivity in 3D

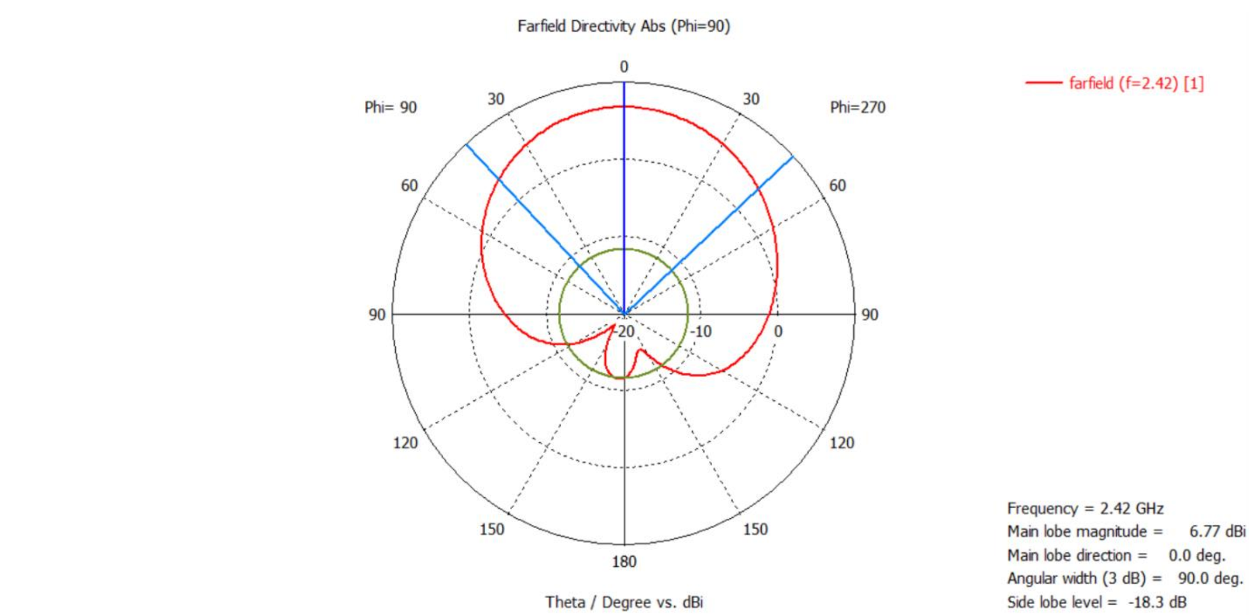


Fig: Farfield Directivity in Polar

Parametric Analysis

- **Effect of Patch Length (L)**
  The influence of patch length on antenna performance was examined by sweeping L between 27.7 mm and 39.27 mm while keeping the dielectric constant (ε_r = 4.0) and substrate height (h = 1.5 mm) fixed. It was found that increasing the patch length caused the resonant frequency to shift downward, as the current path became longer. Conversely, reducing the patch length resulted in an upward shift in frequency. The optimum performance was achieved at L = 30.64 mm, which produced a resonance close to 2.42 GHz with proper impedance matching. This confirms that precise control of L provides an effective way to fine-tune the resonant frequency of a microstrip antenna.

- **Effect of Substrate Thickness (h)**
  The substrate thickness was varied from 1.1 mm to 1.5 mm while keeping L = 30.64 mm and εr = 4.0 constant. Increasing the thickness enhanced the fringing fields, which slightly lowered the resonant frequency and broadened the impedance bandwidth. However, a thicker substrate also increased surface-wave excitation, which can reduce radiation efficiency. Among the tested values, h = 1.5 mm provided the best trade-off between bandwidth and efficiency.

- **Effect of Dielectric Constant (ε_r)**
  The dielectric constant was varied from 2.0 to 6.0, with the patch length adjusted within 29.64 mm to 35.64 mm to maintain resonance near 2.42 GHz. As ε_r increased, the resonant frequency decreased for a fixed patch size, allowing for more compact designs. However, higher ε_r values reduced bandwidth and efficiency due to stronger surface-wave losses. Lower ε_r values provided larger patch sizes but offered better bandwidth and radiation efficiency.

Discussion

The rectangular microstrip patch antenna designed for 2.42 GHz exhibited results that strongly correlated with theoretical predictions. The parametric study confirmed that the patch length is the primary factor determining resonant frequency, substrate height affects impedance bandwidth and efficiency, and dielectric constant influences antenna compactness and performance. The optimized parameters of L = 30.64 mm, h = 1.5 mm, and ε_r = 4.0 achieved stable resonance, sufficient bandwidth, and efficient radiation. This demonstrates the significance of systematic parameter tuning in optimizing microstrip antenna performance for S-band communication applications.

Conclusion

A rectangular microstrip patch antenna was successfully designed, simulated, and analyzed to operate at 2.42 GHz using CST Studio Suite. The final design exhibited efficient radiation performance. The experimental analysis validated the theoretical models and highlighted the importance of optimizing parameters such as patch length, substrate height, and dielectric constant. The results indicate that a properly optimized antenna can provide a compact, efficient, and cost-effective solution suitable for WLAN, radar, and IoT applications operating in the S-band frequency range.